From Calvin Klein to Carrion: Diel Responses Revealed by Camera Traps

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Introduction

- Camera trap surveys are a versatile tool for passively monitoring wildlife and estimating diel activity patterns
- **Goal:** To determine the efficacy of different attractants in estimating activity patterns
- **Objectives:** (1) To compare activity patterns of species with different treatments and (2) to assess interspecific activity patterns

Methodology

- From June to September 2024, 12 cameras were placed on the Winston 8 Ranch property in Lufkin, TX
- The cameras were grouped into three treatment types: No attractant (control), fish, and Calvin Klein Obsession cologne, which mimics civetone, a compound that has been shown to attract felids
- We quantified overlap and diel activity for each species by treatment using kernel density estimation
- Treatment pairs were compared using the overlap coefficient, where values near 1 indicate strong overlap and values near 0 indicate distinct activity patterns (Ridout & Linkie, 2009)



Figure 2: A Turkey vulture visiting a fish treatment station

Results

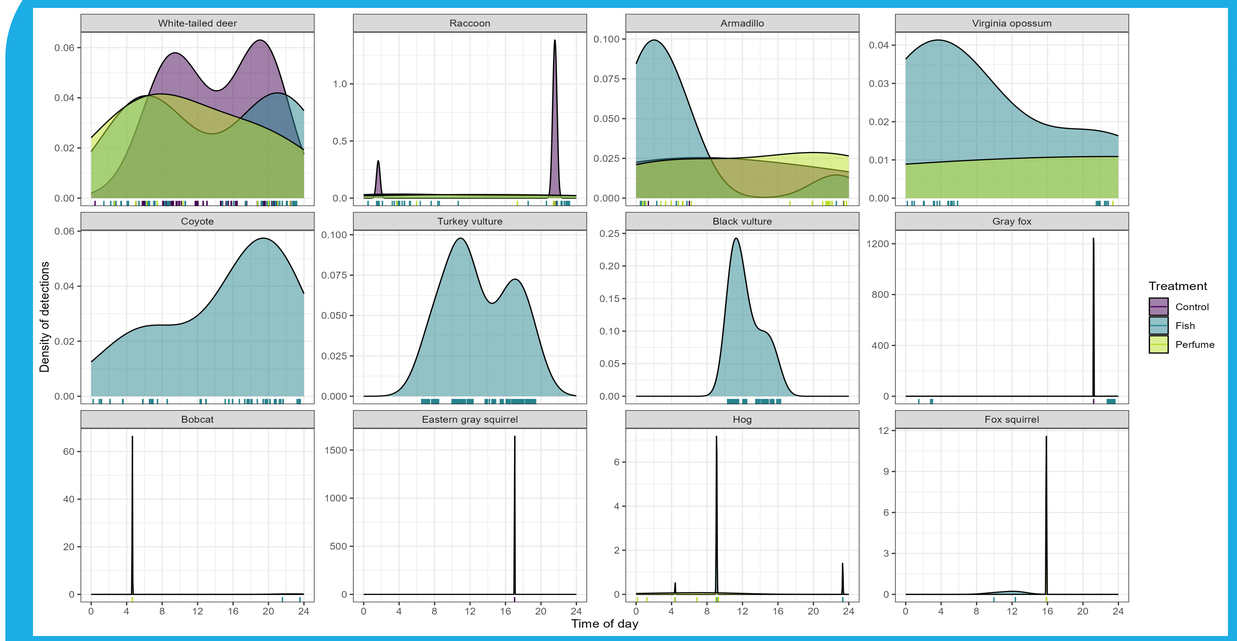


Figure 1: Activity patterns of each species recorded by treatment type and time of day

Species	Comparison	Overlap
White-tailed deer	Control vs. Fish	0.48
White-tailed deer	Control vs. Perfume	0.60
White-tailed deer	Fish vs. Perfume	0.69
Raccoon	Control vs. Fish	0.39
Raccoon	Control vs. Perfume	0.27
Raccoon	Fish vs. Perfume	0.61
Armadillo	Control vs. Fish	0.84
Armadillo	Control vs. Perfume	0.64
Armadillo	Fish vs. Perfume	0.67

Table 1. Activity overlap coefficients for species with sufficient records for overlap comparison

- For white-tailed deer (*Odocoileus virginianus*) and raccoons *(Procyon lotor),* activity overlap was higher between the fish and perfume cameras, with activity peaks at the control cameras
- The highest activity overlap for nine-banded armadillos (*Dasypus novemcinctus*) was between fish and control cameras, with peak activity occurring at the fish stations during early-morning hours
- The coyotes (*Canis latrans*), turkey vultures (*Cathartes aura*), and black vultures (*Coragyps atratus*) were only detected on the fish cameras

Discussion

- The stations with fish head treatment allow for assessing activity patterns and determining activity peaks for different species
- The results showed that white-tailed deer are more active in cameras without bait or lure, suggesting they could perceive risk in those sites (Rocha, 2016)
- Conflict between both species of vulture was observed, as one species tended to leave upon the arrival of the other species and return once they had left
- Understanding how different attractants influence detection rates of different species is necessary for baited camera trap surveys to improve detection of target species that may be elusive, or to better assess the presence of predatory, invasive, or nuisance species



Figure 3: Coyote visiting a fish treatment station

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